

7 On an Argand diagram, the point A represents the complex number z with modulus 2 and argument $\frac{1}{3}\pi$. The point B represents $\frac{1}{z}$.

(a) Sketch an Argand diagram showing the origin O and the points A and B. **[2]**

(b) The point C is such that OACB is a parallelogram. C represents the complex number w .

Determine each of the following.

- The modulus of w , giving your answer in exact form.
- The argument of w , giving your answer correct to **3** significant figures. **[7]**